Can You Automate Expense Tracking in Singapore?

A free tracker built from bank alerts and statements, instead of typing in every payment

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github.com/StarlightsJourney/ugotmonee

Abstract

Most expense trackers in Singapore are manual, because banks don't let personal customers connect apps to their accounts. But banks do send an email for some payments and a statement each month. I built ugotmonee, a free tracker that uses only those two, inside the user's own Google Sheet. It works, with one catch. Over five months of one person's accounts, the emails showed 51% of the money spent but only 27% of the payments. The rest, mostly small card and NETS QR payments, appeared only on the monthly statement. This report explains how to set the tracker up, why the emails miss so much, how the two sources are combined, and the problems I hit along the way.

1 Motivation

There are two usual ways to track spending. One is to connect an app to your bank. In Singapore, banks don't offer this for personal accounts. The other is to type every payment in yourself, which few people keep up.

But your bank already sends you two records of your spending:

- **Email alerts**, seconds after many payments.
- **A statement**, once a month, listing every payment.

ugotmonee reads both and puts every payment into a Google Sheet. It sorts them into categories, draws charts, estimates your balance, and lists your subscriptions. There is no app to install, no account to create, and your data stays in your own Google account.

2 Setting it up

Table 1 lists what you need.

A Google account	Its Gmail must receive your bank's email alerts. Google Sheets is free with it.
Bank email alerts	Turned on in your bank's internet banking, under e-Alerts. Turn on every kind: PayNow, transfers, card payments (set the minimum amount as low as it goes) and NETS QR.
Your statements	The monthly PDFs from internet banking. OCBC and DBS/POSB are supported.
Gemini key	Optional. Only if you want AI to sort merchants the tracker doesn't recognise. Free from Google AI Studio.

Table 1: What you need.

Then:

1. **Make a Google Sheet** and open *Extensions → Apps Script*.
2. **Copy in the three files** from the *apps-script* folder on GitHub. The README shows exactly where each one goes.
3. **Reload the Sheet and click *ugotmonee → Set up*.** Allow access when Google asks. It says the app is unverified because you just made it yourself: click *Advanced*, then *Go to ugotmonee*.
4. **Type your name** in the Settings tab, the way your bank writes it. Money you move between your own accounts then isn't counted as spending.
5. **Import your statements** with *ugotmonee → Import statement (PDF)*.

That's it. New payments appear within five minutes. On the 8th of each month you get an email if last month's statement hasn't been imported yet.

If the menu doesn't appear, or a window says *permission denied*, you are probably signed in to more than one Google account. Open the Sheet in a private window signed in to just one.

3 How much do bank emails show?

For May to September I had both the emails and the statements, so I could check which payments had sent an email. Figure 1 shows the result for each month.

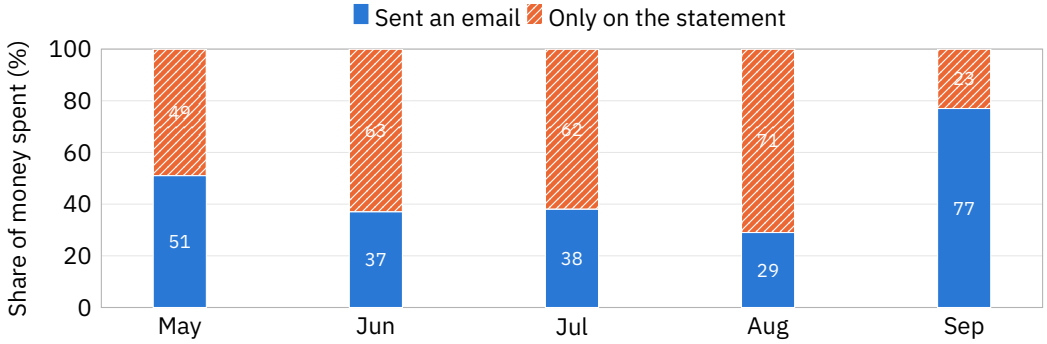

Figure 1: Share of each month's spending that sent an email alert. September is higher only because one large card payment happened to send one.

Overall, emails showed 51% of the money but only 27% of the payments. The missing ones were the small, frequent payments. Table 2 shows what they were.

Kind of payment	Payments	Money	Why there was no email
Card	235	71%	Card alerts only go out above a minimum amount you choose. Bus and train fares are also charged in batches days later.
NETS QR	120	27%	NETS QR alerts are a separate setting, which seems to have been off.
Other	3	2%	Bank fees and a few transfers.

Table 2: Payments that appeared only on the statement, May to September. "Money" is each kind's share of the missing amount.

So a tracker that only reads emails makes every month look much cheaper than it was. Two habits fix most of it: turn on every kind of alert at the lowest amount, and import your statement each month.

4 How it works

Figure 2 shows the parts. Everything runs inside your Google account, using Google Apps Script, Google's free tool for adding scripts to Sheets.

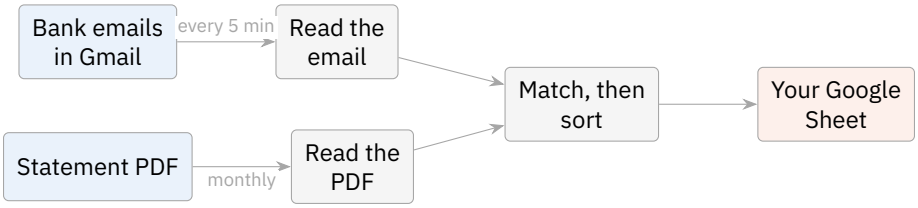

Figure 2: How a payment gets into the Sheet. A payment seen in both an email and a statement becomes one row.

Reading emails. Every five minutes the script searches Gmail for emails from your banks and picks out the date, amount and who was paid. Any email that mentions money but isn't understood goes to an *Unrecognised* tab, which is how new email formats get found.

Reading statements. The PDF is read inside your browser and never uploaded. Each import checks that the opening balance, minus money out, plus money in, equals the closing balance. If it doesn't, the import tells you something was missed.

Matching the two. Most payments appear twice: once in an email and once on the statement. They count as the same payment when the amount matches, the dates are within three days, and the names share a word. When in doubt the tracker keeps both, because a duplicate is easier to notice than a missing payment.

Your balance. The statement's closing balance is exact but only up to month end. After that, the tracker adds the emails since then. Because some payments send no email, you can also type in the balance your bank app shows, and it counts from there.

Subscriptions. The Recurring tab lists payments of about the same amount at a steady interval, like a phone plan, plus known services like streaming. If one stops appearing, it asks whether you cancelled it.

Categories. Each payment gets a category from a built-in list of shops. Change a category once and the tracker remembers it for that shop. Optionally, Gemini sorts shops the list doesn't know. It sees only the shop's name, never amounts, dates or account numbers.

5 Problems I hit

Table 3 lists the problems that took the most time to solve. Most aren't in Google's documentation.

Problem	What happened	Fix
Reading the email	Every email failed to read	Gmail returns the email as raw bytes, not text as its documentation says. Accept both.
Too many emails at once	On the first run, Gmail's per-minute limit was hit	Stop when the limit is hit, and carry on from there five minutes later.
Six-minute limit	Google stops any script after six minutes	Read emails for three minutes, then save everything in one go.
New permissions	After an update, syncing quietly stopped	Click Set up again to approve the new permission.
Several Google accounts	"Permission denied" and a missing menu	A known Google bug. Use a private window with one account.
Wrong monthly totals	Every month showed the same total	One spreadsheet formula doesn't work across a list of months. Use a different one.
Silent failures	Scheduled runs failed without any message	The Settings tab now shows when the last sync ran and how it went.
Bank wording	"FAST Payment / Receipt" on a statement never matched its email	When the statement names nobody, match on amount, account and day.

Table 3: The main problems, what happened, and the fix.

6 Privacy

Your payments never leave your Google account, unless you turn on Gemini, which sees only shop names. The code is public, so it contains no real data. Test examples use made-up names and numbers, and the repository refuses any commit that contains a statement, a password or key, or an NRIC or phone number.

7 What's next

This is a first version. Right after this study I turned on card alerts at the lowest amount and switched on NETS QR alerts. The tracker already reads both kinds of email, so from November most of the card and NETS QR payments that used to appear only on statements should arrive by email within minutes instead of a month later. The next version of this report will measure how much that closes the gap.

8 Limitations

I tested one person's accounts at two banks. Other banks will send emails the tracker hasn't seen yet, and these will land in the Unrecognised tab. Payments your alert settings leave out only appear when you import the statement. E-wallet balances aren't tracked, which is fine as long as you keep the wallet empty.

A Email formats

These are the email formats the tracker reads. All examples are made up.

Bank	How to recognise it	Notes
OCBC	PayNow to a business: “has been made to X using their Unique Entity Number”	Times can look like “16:51 PM”.
OCBC	PayNow to a person: “using his/her Mobile”	Names can contain extra spaces.
OCBC	PayNow failed: “could not be made to”	Skipped, since no money moved.
OCBC	Transfer to another bank: “request to make the following transfer”	Time written with a dot, “11.14 AM”.
OCBC	NETS QR: “The following NETS QR payment has been made”	The account name leaves out “OCBC”.
OCBC	ATM: “was withdrawn from your account at an OCBC ATM”	Sorted as cash.
OCBC	Card: “was charged … to your card”	Only above your card alert amount.
OCBC	Red packet (e-Packet)	Sorted as gifts.
DBS	PayLah!: “We refer to your PayLah! … dated”	No year given; worked out from when the email arrived.
DBS	Money received: “You have received SGD … via FAST transfer”	Money coming in. OCBC sends no email for this.

Table 4: Email formats from OCBC and DBS/POSB.